

Memristor-based Deep Spiking Neural Network with a Computing-In-Memory Architecture

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Abstract

Spiking Neural Networks (SNNs) are artificial neural network models that show significant advantages in terms of power and energy when realizing deep learning applications. However, the data-intensive nature of machine learning applications imposes a challenging problem to neural network implementations in terms of latency, energy efficiency and memory bottleneck. Therefore, we introduce a scalable deep SNN to address the problem of latency and energy efficiency. We integrate a Computing-In-Memory (CIM) architecture built with a fabricated memristor crossbar array to reduce the memory bandwidth in vector-matrix multiplication, a key operation in deep learning. By applying an inter-spike interval (ISI) encoding scheme to the input signals, we demonstrate the spatiotemporal information processing capability of our designed architecture. The memristor crossbar array has an enhanced heat dissipation layer that reduces the resistance variation of the memristors by $\sim 30\%$. We further develop a time-to-first-spike (TTFS) method to classify the outputs. The designed circuits and architecture can achieve very high accuracies with both digit recognition and the MNIST dataset. Our architecture can classify handwritten digits while consuming merely 2.9mW of power with an inference speed of 2 μ s/image. Only 2.51pJ of energy per synaptic connection makes it suitable to apply in deep learning accelerators.

Keywords

Spiking Neural Network, Computing-In-Memory, LIF Neuron, memristor, spatiotemporal architecture.

1. Introduction

Deep learning has shown excellent results in the field of large data analysis [1]. Deep neural networks (DNNs) can be applied to speech recognition, digit recognition, pattern classification and natural language processing. The efficiency of these neural networks can be improved significantly by using Spiking Neural Networks (SNNs). Compared to the traditional neural networks, SNNs mimic biological neurons more closely because of their spiking nature; once a threshold is exceeded, a spike is fired, and the information is conveyed via the specific time of the spikes or the ordering thereof [2]. Furthermore, the binary nature of the spikes in SNNs contributes to efficiency in terms of both power and energy. They carry a spatiotemporal information processing capability where the spatial nature is a result of neurons connected locally to each other and the temporal nature comes from the spiking events that occur at specific times [3]. SNNs are therefore said to be the third generation of artificial neural networks [3].

A number of encoding schemes have been developed to convert the data into spike events. The two main encoding

schemes are rate encoding and temporal encoding. Rate encoding is where the data is encoded into the frequency of the spikes and temporal encoding is where the data is encoded into the exact timing of the spikes [4], [5]. In temporal encoding scheme some common ones are time-to-first-spike (TTFS), phase and burst encoding [6], [7], [8]. Temporal encoding is more energy efficient compared to rate encoding because of the fewer spike events generated [5]. In this work, we use a type of temporal encoding scheme called the inter-spike interval (ISI) encoding scheme to encode the data into spike events. The key idea of this ISI encoding scheme is that it has higher information density compared to its TTFS counterpart since it carries information in both the timing of the spikes as well as the distance between the spikes based on the intensity of the information.

Compared to the traditional CMOS memory technologies that include static random access memory (SRAM), dynamic random access memory (DRAM) and flash, emerging non-volatile memory technologies (eNVMs) have shown to emulate biological neurons and synapses more closely which is invaluable to in-memory computations for neuromorphic hardware. A particular eNVM called resistive random access memory (ReRAM) or memristor has received wide attention due to its scalability, compatibility with CMOS technology, analog conductance modulation and low power consumption [9]. Using ReRAMs for in-memory computation operations also replaces the need for power-hungry and area-hungry analog-to-digital and digital-to-analog converters.

In this work, we developed a novel computing-in-memory (CIM)-based SNN architecture with the fabricated memristor crossbar [10] that has a unique heat dissipation capability to reduce the resistance variation by almost 30%. The memristor also has a competitive cycle-to-cycle variation of only 4%. The high on and off resistance ratio provides a stable high and low state that is favorable for binarized neural network, as well as minimizes leakage currents in memristor crossbars, which is a key issue for memristors. The main contributions of this paper are listed as follows:

- A novel scalable deep neural network is designed using a CIM architecture with memristor-based crossbars.
- A spatiotemporal information processing capability is demonstrated where each row of pixels can be taken in for one clock cycle.
- Our design incorporates a novel ISI encoding scheme for higher information density and uses it as an interface between layers.
- The design uses a memristor model for the construction of the crossbars that allows for vector-matrix multiplications, followed by a current compensator stage that adds variation to compensate for both binary weights as well as leakage current.

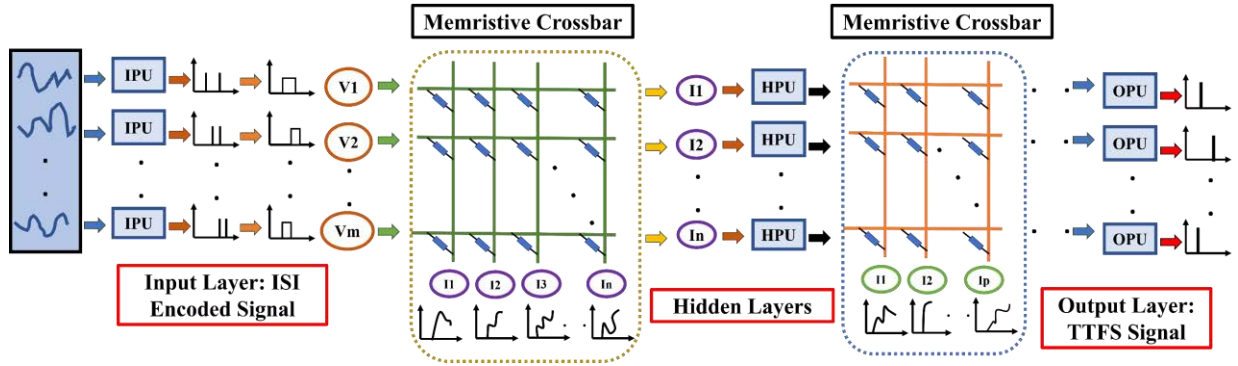


Figure 1: Architecture of the designed CIM-based SNN

- The accuracy of the design is evaluated on hardware using the pixels from images of digits 0-9 with the TTFS classification scheme.
- We developed a large-scale CIM-based SNN to evaluate our design performance by the MNIST handwritten digits and our results demonstrate significantly higher accuracy with the ISI encoding scheme compared to its TTFS and rate counterpart.

2. Background and Related Work

2.1. Encoding Schemes

In principle, information needs to be encoded into a spike train before it can be fed into the SNN architecture. Significant research has been done in the area of neural encoding schemes since it is essential in conveying accurate information while consuming low power and latency [7], [8], [11]. Thus far, temporal and rate encoding are the two commonly used neural encoding schemes in literature [12], [13]. Despite its robustness and simple implementation characteristics, rate encoding suffers from latency and significant energy consumption arising from the large number of spikes generated [14], [15]. Temporal encoding schemes on the other hand have shown to emulate biological neural systems more closely with their ability to capture the temporal aspects of the spikes while significantly reducing the number of spikes generated [13]. Many types of temporal encoding schemes including phase and burst encoding schemes have been applied to deep SNNs and

have shown significantly higher efficiency compared to DNNs [7], [8]. Nevertheless, the issue of large number of spike generation prevents them from reaching the desired performance.

In this work, we demonstrate a type of temporal encoding scheme for the input layer, the ISI encoding scheme, that has the capacity of encoding information in two dimensions [4], [16]. To preserve the information density, we use the ISI encoding in the input layer and for the output classification layer we use another type of temporal encoding, the TTFS encoding scheme for its high energy and power efficiency by generating only one spike. The following section discusses the details of our circuits and architecture design.

2.2. Computing-In-Memory

Artificial intelligence applications have a memory-intensive nature which makes it difficult for them to run on traditional computer architectures [17]. Neural network operations require a large amount of data movement to and from memory because of the number of vector-matrix multiplication operations and fetching of weights that are stored in the memory. The performance of the system is limited by the speed of the movement of data to and from memory. Furthermore, a significant amount of energy is spent due to the physical movement of data that arises from interconnect parasitics, which will exacerbate over time as technology scales [18]. A memristor crossbar provides a

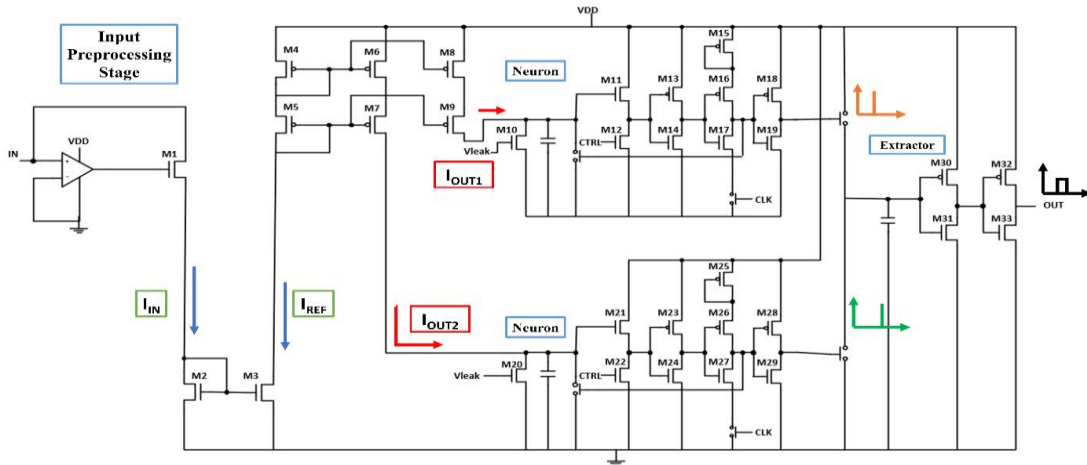


Figure 2: Input processing unit

solution to this problem by allowing large amount of vector-matrix multiplication in-memory, consuming extremely low area and low power [17]. Therefore, in this work we adopt a CIM architecture with a fabricated memristor crossbar and integrate it with our SNN architecture.

3. ISI-Based SNN Circuits and Architecture

For our designed architecture portrayed in Figure 1, we combine both the SNN configuration and the CIM structure. Our deep SNN architecture is constructed with four major blocks, an input layer consisting of the temporal ISI encoder, a fabricated memristor crossbar array for the vector-matrix multiplications, a hidden layer for propagating the signal through the layers and an output layer consisting of a TTFS decoder.

3.1 Input Layer

The higher information density in the ISI scheme is achieved by encoding the information in both the time of the spikes and the time correlation between the spikes [19]. Despite the higher information density in the ISI encoding scheme, it still experiences the drawbacks of latency and large number of spike generation. In our version of the ISI encoding scheme, we address the issue of latency and the considerable amount of spike generation suffered by temporal encoding scheme by generating two spikes and capturing the time between them. The detailed circuit design of our ISI encoding scheme is illustrated in Figure 2.

Our designed ISI encoding circuit consists of three major modules, the input preprocessing stage, the neuron stage and finally, the extractor stage. This input preprocessing module utilizes the current-to-current conversion method with the help of the conventional transconductance amplifier. The current mirror array in the input preprocessing stage detects the variation of the input signal I_{in} and linearly generates the corresponding tuned current signals I_{out1} and I_{out2} given by the following equations,

$$I_{out1} = m \cdot I_{in} \quad (1)$$

$$I_{out2} = n \cdot I_{in}, \quad (2)$$

where m and n are scaling factors to add additional weights to the system. The two tuned current signals are applied to two leaky-integrate-and-fire (LIF) neuron model that uses the capacitor-sensing approach to generate the spikes. The incoming input excitation current can be written using the equation shown below:

$$I = C \cdot \left(\frac{dV}{dt} \right) + I_{leak} \quad (3)$$

where C is the membrane capacitor for sensing and I_{leak} is the leakage current from the neuron. The sensing capacitor tracks the current signal applied continuously and charges up to its potential. Meanwhile, an external clock signal CLK controls the frequency of the spikes. When the voltage across the sensing capacitor exceeds the threshold set by the $CTRL$ signal, the corresponding cascaded inverters ($M_{13, 14}$ and $M_{18, 19}$) and ($M_{23, 24}$ and $M_{28, 29}$) generate an output spike. Once a spike is generated, the sensing capacitor is reset by the positive feedback loop from the transistors ($M_{16, 17}$ and $M_{26, 27}$).

Following the operation of the LIF neurons, when the two tuned current signals are applied to the two LIF neurons, two spikes are generated for each clock cycle as portrayed in Figure 2. With I_{out1} having a higher current value than I_{out2} , the spike from the top LIF neuron appears sooner than the bottom LIF neuron. These two spikes are applied to the extractor stage where the capacitor sensing technique is used to generate a pulse signal, capturing the time between the spikes. These pulse signals can then be applied to the memristor crossbar stage.

3.2. Memristive Based CIM Layer

To compute vector-matrix multiplications using the proposed CIM architecture, the input vectors are mapped as voltages while the weight matrices are mapped as conductance values in the memristor crossbar. The horizontal and vertical lines are the word-lines and bit-lines, respectively. The output accumulated currents from the bit-lines in the crossbar can be described by the equation:

$$I_j = \sum_{i=0}^n G_{ij} V_i, \quad (4)$$

where G is the conductance of the memristor and the subscripts i and j are the rows and columns of the crossbar, respectively. Because of this crossbar structure, the memristor can be used to implement large-scale neural networks due to its high density and non-volatility. Furthermore, its nanoscale structure, low operating voltage and power consumption make it suitable to apply it in vector-matrix multiplications [20].

A two terminal memristor device is used to build the crossbar as discussed in the preceding section. Memristors have a metal-insulator-metal structure that operate by changing their resistance from a high resistance (HRS) to a low resistance (LRS) that occur due to voltage or current applied to trigger the formation and rupture of the conductive filament [20]. This conductive filament is in between the top and the bottom electrodes that creates a path for the memristor to conduct current [10], [21]. In this architecture design, a unique memristor device is used that has a distinctive heat dissipation capability. One of the major issues of memristors is low reliability and accuracy that comes from the variation of resistances inside the device. This resistance variation issue has been extensively

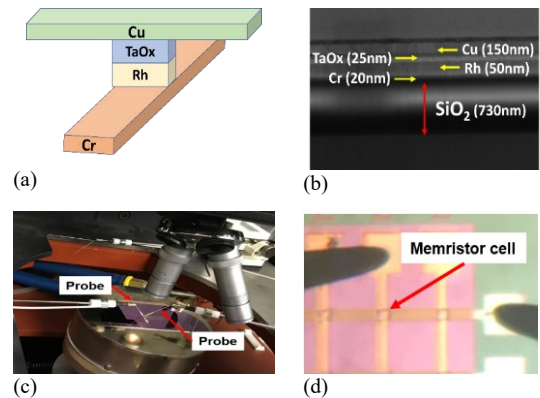


Figure 3: (a) Memristor metal-insulator-metal structure. (b) Fabricated memristor structure. (c) Memristor wafer with attached probes. (d) Zoomed-in view of our 5x5 memristor

studied and it has been found that a major factor contributing to the device variation is the heat dissipated during the destruction and creation of the conductive filament [10], [21]. This problem was resolved by this memristor device by introducing an additional heat dissipation layer inside the metal electrodes to compensate for resistance switching variation by almost 30% [10], [21]. This memristor has a Cu/TaOx/Rh/Cr configuration with a cycle-to-cycle variation of only 4%. For the active anode, Copper was used because of its medium activation energy that allows it to release ions easily. Rhodium was used to make the inert cathode due to its compatibility with CMOS technology and for the oxide layer, tantalum oxide was used. The additional metal dissipation layer of Chromium in this memristor was able to reduce the effects that arise from heating and therefore can reduce the resistance variation. To reduce the impacts of the Joules heating effect, multiple materials were tested including Rh, Cr, Pt, Ti and Cu and it was concluded that replacing the Ti glue layer with Cr which has the most thermal conductivity was a viable solution. To write the weights onto the memristor crossbar, write pulses of the same amplitude can be applied (1V, 10ns).

The fabricated 5x5 memristor crossbar is shown in Figure 3 where the memristor is in the cross-section of the top and bottom nanowires. The memristors in the crossbar also have a reduced cycle-to-cycle variation of only 4% that further improved the reliability of the device. The high on and off ratio of this memristor of $1\text{M}\Omega$ to $980\text{M}\Omega$ also mitigates some of the problems that arise from leakage currents in the crossbar and allows it to have stable on and off states.

3.3. Hidden Layer

The designed SNN architecture from Figure 1 is a scalable network. Our architecture allows for multiple hidden layers to be incorporated. To add the hidden layers to the network, the output current from the columns of the memristor crossbar array have to be post-processed. Therefore, these current signals are amplified using the circuit design from Figure 2. Using this circuit, the leakage current arising from the nanowires of the crossbar can also be compensated with the amplification of the output current. In the hidden layer the output is converted from the

accumulated current signal to an ISI encoded signal using the similar mechanism as the input layer, where the signal is initially converted to two spikes and then to a pulse signal. This introduces scalability to our structure since the ISI encoding stage can be reproduced in the hidden layer with the desired current amplification. Thus, the architecture maintains the propagation of spikes between the memristor crossbar stages, allowing for fast and energy efficient transmission of the signals.

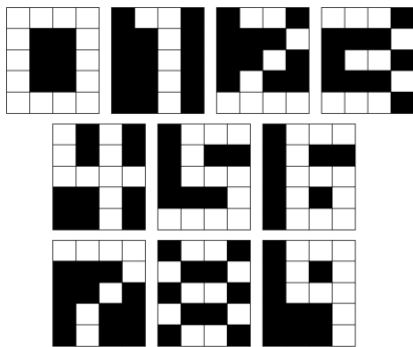
3.4. Output Layer

The input to the output layer is the bit-line current from the columns of the memristor crossbar. A TTFS decoder is adopted in the output layer to carry out the classification. TTFS is the most energy and power efficient encoding scheme since only one spike is generated per clock cycle [6]. For the input encoding scheme, we use the ISI method to incorporate the higher information density as it passes through our entire network but since the output will only separate the different classes, we utilize the TTFS method. Before the output current can be used to carry out the classification process, it needs to be amplified in order to allow the neurons to integrate sufficient current necessary to fire. Similar to the current-to-current conversion module from the input stage, the current signal is copied and linearly converted to an amplified output current based on the scaling factor. After the current is scaled to a higher value, it is then applied to one LIF neuron controlled by the same external clock as the entire architecture. For one clock cycle the input is fed parallelly to the entire network and in the output stage each LIF neuron will generate a spike to be used for classification.

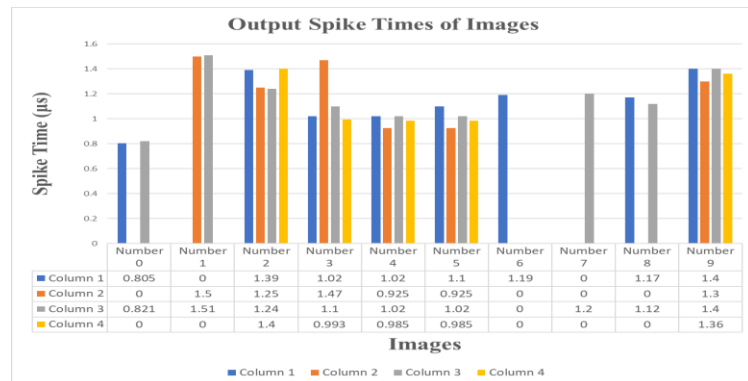
4. Simulation Results

4.1. Training and Classification of 5x4 Images

To evaluate the SNN design, we use the images 0 to 9 as inputs to the network shown in Figure 4(a). A three-layer neural network is used to process the images. Since a small-scale neural network is used with an output layer of 4x4 crossbar, the weights are trained to identify the digits 0 to 3. Each entire row of pixels is taken in for one clock cycle and applied to a 5x5 crossbar. The crossbar has the trained weights. The input signals are preprocessed first by applying them to the weighted input crossbar to add some variation to the signal. From the columns of the first crossbar there are three levels of output currents, $0\mu\text{A}$, $1.8\mu\text{A}$ and $3.6\mu\text{A}$ that



(a)



(b)

Figure 4: (a) Images used for the 5x4 classification. (b) Output spike times of images from the classification.

are the results of the multiply and accumulate operations. These output currents are then applied to the input processing units.

When the pulse signals from the input stage are applied to the memristor crossbar, the input signals or the vectors get multiplied with the conductance values stored in the memristor crossbar and produces an output column current. The hidden stage that follows can be replicated from the input stage which makes the SNN structure a scalable architecture that can be used for larger data processing. The output currents from the crossbar in the final stage are then applied to the output stage where the current is amplified and applied to an LIF neuron stage to generate the output spike. Even though the output crossbar is only 4x4 and therefore is able to classify 4 digits, the rest of the outputs from the digits 4-9 are tabulated in Figure 4(b), to show that the spiking pattern for each neuron is also different, which can be used to classify the images. A clock frequency of 0.5MHz was used. Each image is processed in one clock cycle which is 2μs per image using a supply voltage of 1.8V.

The evaluation of the area and power for each processing unit was done using Cadence Virtuoso using a supply voltage of 1.8V demonstrated in Figure 5. For each processing unit the LIF neuron only consumes 4.37μW of power. The majority portion of the power is consumed by the current mirror stage which consists of an opamp and the entire current mirror structure and consumes 115μW of power. The extractor stage which produces the ISI pulse consumes the least amount of power. The power consumed by each input processing unit is 123μW. The total power consumed by the entire SNN architecture is only 2.9mW. From the area breakdown of the input processing unit, it can be seen that a major portion of the area, 55%, is occupied by the current mirror stage and the extractor unit and the LIF unit consumes only 19% and 26% of the area respectively.

A comparison with the state-of-the-art neuromorphic chips from Table 1 shows that the designed architecture has a competitive power consumption of only 2.9mW for the entire network. It also has a low operating voltage and a Compute-In-Memory mode and uses LIF neurons which are more biologically realistic models. The average energy consumption per synaptic connection for our design is only 2.51pJ. Since the designed architecture was simulated in small-scale, some of the performance parameters of the system is lower.

4.2. Comparison with Various Encoding Schemes

To test a large-scale network, we developed a three-layer neural network that incorporates the ISI encoding

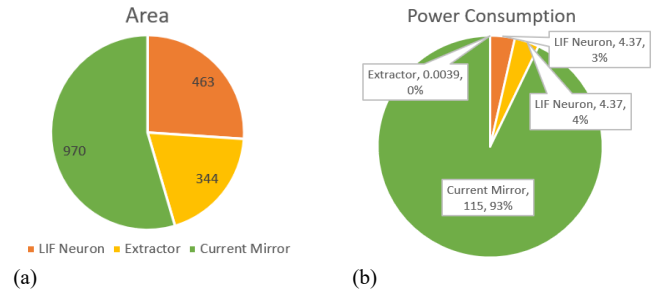


Figure 5: (a) Area breakdown. (b) Power breakdown.

scheme and the memristor crossbar structure. This model was used to test the accuracy against the MNIST dataset between the different encoding schemes, namely rate, TTFS and ISI encoding. The large-scale three-layer SNN model is inspired by [26] with an encoding stage where the inputs are initially encoded into either one of rate, TTFS, or ISI. After the encoding stage the inputs are applied to the memristor crossbar layer that performs the matrix multiplication. Here the memristor weights are trained via the ADAM optimizer. After the crossbar layer the Batch Normalization layer is added, which mimics the current mirror stage. In this stage column output currents are converted to voltages clamped between two pre-specified design parameters. After this layer, a set of LIF neuron layer is added, which controls the dynamics of the first layer, generating a sequence of spikes. It was observed that the use of the ISI encoding scheme provides the highest accuracy of 90% on the designed SNN architecture compared to the rate encoding which is 83% and the TTFS encoding which is 85% [26].

5. Conclusion

SNNs provide significant advantage in terms of power and energy consumption by conveying information through spikes. A novel ISI encoding scheme is used for the input which has higher information density compared to its TTFS and rate counterpart. A memristor crossbar with a heat dissipation capability is used to carry out the vector-matrix multiplication process. The designed CIM-based SNN architecture includes the detailed circuit design for the ISI encoded input stage, the hidden stage, and the output stage with a TTFS method. Furthermore, the design has a spatiotemporal information processing capability since it can take in an entire row of pixels over one clock cycle. This gives the SNN a very high inference speed of 2μs/image. The whole network that was used in handwritten digit recognition has a power consumption of only 2.9mW. Our

Table 1: Performance Comparison with the State-of-the-Art Neuromorphic Chips

	[22]	[23]	[24]	[25]	This Work
Technology	180nm	65nm	130nm	130nm	180nm
Algorithm	CNN	CNN	RBM	MLP	SNN
Memory Cell	SRAM	SRAM	ReRAM	ReRAM	ReRAM
Memory Mode	-	-	CIM	CIM	CIM
Weight Precision	-	-	1-bit	3-bit signed	1-bit
Neuron Type	IF	-	IF	-	LIF
Supply Voltage	1.8V	1V	1.8V	5V	1.8V
Latency	15.4ns	-	-	51.1ns	334ns
Power Consumption	4mW	278mW	2.2mW	-	2.9mW

designed architecture achieves a 90% accuracy with the MNIST dataset using an ISI encoding scheme which is significantly higher than TTFS and rate encoding.

6. Acknowledgment

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